

D660

Instructional Technology and Online Pedagogy

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A middle school science teacher wants students to understand why the moon has phases. Her school recently won a technology grant, and she enjoys trying new apps on weekends. Which choice best reflects TPACK?

- A. A typing game to build keyboard skills first.
- B. The newest app from the grant list, whatever its topic.
- C. An interactive simulation where students move the moon around Earth and observe the lit portion from Earth.
- D. Slides of the same textbook paragraph, projected on a screen.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

Students used to write a paper report on local history. Now, working on laptops in a quiet library, they interview elders by video, build a shared map with recordings, and publish it for the town museum's website. Which SAMR level fits best?

- A. Augmentation
- B. It does not count, because the work is done in a library.
- C. Redefinition
- D. Substitution

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A teacher wants a SAMR substitution-level example. Which one fits?

- A. Students create a podcast series for listeners in another country.
- B. Students co-author a story in a shared document and comment on each other's drafts.
- C. Students type a spelling list in a word processor instead of writing it by hand.
- D. Any task done on a computer is substitution.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

An online high school course has students in three time zones, many of whom work evening jobs. The course platform has a new purple theme. The teacher wants every student to write thoughtful, researched responses to a debate question. Which format fits best?

- A. A live video debate at 7 p.m. in one time zone.
- B. An asynchronous discussion board with a deadline and reply expectations.
- C. Change the platform theme so students log in more.
- D. A live session is always better, so hold one anyway.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

An online teacher needs to walk a small group through a tricky lab procedure, answer questions as they arise, and correct mistakes on the spot. Which format fits this goal?

- A. An end-of-unit online test.
- B. An asynchronous discussion board.
- C. A recorded video with no way to ask questions.
- D. A synchronous video session with the small group.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A teacher is making a 5-minute instructional video on the parts of a cell. He records in his kitchen and uses a free editing program. He plans to add background music, a spinning logo, and the full script printed on screen while he narrates. What change would most improve learning?

- A. Remove the music and logo and show labeled diagrams while he narrates, without the full script on screen.
- B. Keep the full script on screen so students can read along.
- C. Add more animations so students stay interested.
- D. Rerecord in a studio.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A teacher is posting a lecture video and a diagram of the digestive system to an online course. The course has 32 students and the video was recorded on a new phone. Which step best makes the content accessible?

- A. Wait until a student asks for captions.
- B. Record in higher resolution.
- C. Use alt text that says "image."
- D. Add accurate captions to the video and descriptive alt text to the diagram before posting.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Students find a website about a health topic with a .org address, a professional design, and an About page saying it is independent. The site loads quickly and has many photos. What should students do to check its credibility?

- A. Open new tabs and search what reliable outside sources say about who runs the site.
- B. Trust it because it ends in .org.
- C. Trust it because it looks professional and loads fast.
- D. Read the About page closely.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student quickly decided a site was unreliable after searching its sponsor and finding news reports that it is funded by a company selling the product it reviews. Which skill did the student use?

- A. Lateral reading
- B. Vertical reading
- C. Skimming for keywords
- D. Citation formatting

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A fourth-grade teacher finds a free, popular quiz app. It asks for each student's full name, birthday, and email to create accounts. Other teachers on social media love it, and it has bright graphics. What should the teacher do first?

- A.** Use it, since free apps do not collect data.
- B.** Check whether the district has approved the app and reviewed its privacy terms before using it.
- C.** Start using it, since many teachers already do.
- D.** Have students create accounts with their own personal emails.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A teacher wants to show the school board how average attendance changed each month from August to May. The data is in a shared spreadsheet the office manager updates, and the board meets in the library. Which display fits best?

- A.** A pie chart with one slice for each month.
- B.** A single number for the year's average.
- C.** A table of every student's daily attendance.
- D.** A line graph with months on the horizontal axis.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A pie chart in a report is labeled “Share of this year's technology budget by category.” What question does this chart answer?

- A.** Which school spent the most on technology?
- B.** What portion of the total budget went to each category?
- C.** Every question about the budget.
- D.** How did the budget change from year to year?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A middle school science teacher wants students to understand why the moon has phases. ~~Her school recently won a technology grant, and she enjoys trying new apps on weekends.~~ Which choice best reflects TPACK?

Target word: tool chosen through content and pedagogy **Decides:** The tool must help students with the content idea.

A	Adjacent category	A tool skill unrelated to the content goal.
B	Camouflage	Chooses by novelty and funding, not by content.
C	Correct	Tool, method, and content align to build the concept.
D	Dropped element	Uses a tool but drops any change in how students engage the content.

2. Answer C Concept 2

Students used to write a paper report on local history. Now, ~~working on laptops in a quiet library,~~ they interview elders by video, build a shared map with recordings, and publish it for the town museum's website. Which SAMR level fits best?

Target word: SAMR level **Decides:** The task could not exist without the technology.

A	Confusion set	Augmentation adds small improvements, like spell check, to the same task.
B	Camouflage	Location does not decide the SAMR level.
C	Correct	Creating and publishing a new multimedia product for a real audience is redefinition.
D	Confusion set	Substitution is a direct swap with no functional change, like typing the same report.

3. Answer C Concept 2 · transfer

A teacher wants a SAMR substitution-level example. Which one fits?

Target word: reverse: pick the task at a given level **Decides:** Substitution = same task, tool swap, no functional change.

A	Adjacent category	That is redefinition.
B	Inverted operation	Real-time collaboration changes the task, which is modification.
C	Correct	Same task, new tool, no added function.
D	Absolute language	The level depends on how the task changes, not the device.

4. Answer B Concept 3

An online high school course has students in three time zones, many of whom work evening jobs. ~~The course platform has a new purple theme.~~ The teacher wants every student to write thoughtful, researched responses to a debate question. Which format fits best?

Target word: sync vs. async for the goal **Decides:** Thoughtful, researched responses and varied schedules.

A	Inverted operation	Real-time format excludes working students and limits reflection.
B	Correct	Allows time to research and fits different schedules.
C	Camouflage	The theme does not affect the format choice.
D	Absolute language	The best mode depends on the goal and learners.

5. Answer D Concept 3 · transfer

An online teacher needs to walk a small group through a tricky lab procedure, answer questions as they arise, and correct mistakes on the spot. Which format fits this goal?

Target word: reverse: goal calls for the other mode **Decides:** Immediate back-and-forth feedback is needed.

A	Adjacent category	Assessment, not live guided instruction.
B	Inverted operation	Delayed replies cannot fix mistakes in the moment.
C	Dropped element	Drops the interaction the goal needs.
D	Correct	Real-time interaction allows instant questions and correction.

6. Answer A Concept 4

A teacher is making a 5-minute instructional video on the parts of a cell. ~~He records in his kitchen and uses a free editing program.~~ He plans to add background music, a spinning logo, and the full script printed on screen while he narrates. What change would most improve learning?

Target word: multimedia principle **Decides:** Extra material and printed narration overload learners.

A	Correct	Applies coherence and redundancy principles.
B	Confusion set	Same words printed and spoken compete for attention.
C	Inverted operation	More extraneous material increases load.
D	Camouflage	Where it is recorded does not fix the design problem.

7. Answer D Concept 5

A teacher is posting a lecture video and a diagram of the digestive system to an online course. ~~The course has 32 students and the video was recorded on a new phone.~~ Which step best makes the content accessible?

Target word: accessible content at posting **Decides:** Captions and alt text built in from the start.

A	Inverted operation	Reactive accessibility leaves barriers in place.
B	Camouflage	Video quality does not address access needs.
C	Dropped element	Alt text must describe the meaningful content.
D	Correct	Makes both media usable for all learners from the start.

8. Answer A Concept 6

Students find a website about a health topic with a .org address, a professional design, and an About page saying it is independent. ~~The site loads quickly and has many photos.~~ What should students do to check its credibility?

Target word: lateral reading **Decides:** Check what other sources say about the site.

A	Correct	Lateral reading checks a source from the outside.
B	Absolute language	Anyone can register a .org address.
C	Camouflage	Design and speed say nothing about accuracy.
D	Adjacent category	A site's self-description is not independent evidence.

9. Answer A Concept 6 · transfer

A student quickly decided a site was unreliable after searching its sponsor and finding news reports that it is funded by a company selling the product it reviews. Which skill did the student use?

Target word: reverse: name the skill from the action **Decides:** Leaving the site to check it elsewhere.

A	Correct	Checking what others say about a source is lateral reading.
B	Inverted operation	Vertical reading stays on the page and examines it from within.
C	Adjacent category	A reading strategy, not a credibility check.
D	Confusion set	Formatting sources does not evaluate them.

10. Answer B Concept 7

A fourth-grade teacher finds a free, popular quiz app. It asks for each student's full name, birthday, and email to create accounts. ~~Other teachers on social media love it, and it has bright graphics.~~ What should the teacher do first?

Target word: privacy before adoption **Decides:** The app collects student personal data.

A	Absolute language	Free apps often collect data.
B	Correct	Student data rules (FERPA, COPPA, district policy) come first.
C	Camouflage	Popularity does not show legal or policy compliance.
D	Inverted operation	Increases exposure of student data.

11. Answer D Concept 8

A teacher wants to show the school board how average attendance changed each month from August to May. The data is in a shared spreadsheet ~~the office manager updates, and the board meets in the library.~~ Which display fits best?

Target word: chart type for the question **Decides:** Change over time.

A	Confusion set	Pie charts show parts of a whole, not change over time.
B	Dropped element	Drops the month-by-month change.
C	Adjacent category	Raw detail hides the trend the board needs.
D	Correct	Line graphs show trends over time.

12. Answer B Concept 8 · transfer

A pie chart in a report is labeled "Share of this year's technology budget by category." What question does this chart answer?

Target word: reverse: from display to question **Decides:** Pie charts show parts of one whole.

A	Adjacent category	Comparing schools needs a bar chart of categories.
B	Correct	Parts of a single whole.
C	Absolute language	Each display answers a specific kind of question.
D	Confusion set	That needs a line graph over time.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	7	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	6	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	TPACK: content and pedagogy before tools	Good technology integration blends content knowledge, teaching method, and tool; the tool is chosen for what it lets students do with the content.	Starting with the newest tool and looking for a place to use it.	A teacher choosing among tools for a specific learning goal.
2	SAMR levels	Substitution and augmentation enhance a task; modification and redefinition transform it into something new or impossible without the technology.	Calling any device use transformation.	Describing what the technology changes about a task.
3	Synchronous vs. asynchronous	Synchronous suits real-time interaction and quick feedback; asynchronous suits reflection, flexibility, and different time zones or schedules.	Assuming live sessions are always better online teaching.	A goal paired with learners' schedules or a need for reflection.
4	Multimedia principles	Learning improves when extra material is cut (coherence) and narration is paired with graphics instead of the same words printed on screen (redundancy).	Adding music, decorative images, and full on-screen text to be more engaging.	A video or slide design choice.
5	Accessible digital content	Captions for video, alt text for meaningful images, and structured headings make content usable for all learners from the start.	Adding accessibility only when a student requests it.	Posting media or documents to an online course.
6	Digital literacy: lateral reading	To judge an online source, leave the page and check what other reliable sources say about it.	Judging credibility from a site's look, .org domain, or its own About page.	Students evaluating an unfamiliar website.
7	Student data privacy	Before adopting an online tool that collects student data, check it against district approval and privacy laws such as FERPA and COPPA.	Assuming a free, popular app is safe to use with student accounts.	A new app that asks for student names or emails.
8	Choosing the right data display	Line graphs show change over time, bar charts compare categories, and pie charts show parts of one whole.	Using a pie chart for change over time.	What the data is meant to show.

Course scope (public description): Instructional Technology and Online Pedagogy is an innovative course designed to equip students with the skills to effectively integrate technology in their teaching practices. The course also covers best practices for online pedagogy, assessment and feedback, collaborative learning, and the use of multimedia and interactive elements to enhance learning experiences. With a focus on practical application, students will leave the course ready to create and facilitate compelling, high-quality online learning experiences that meet the needs of today's diverse learners. This course also provides a foundation for supporting digital literacy in K-12 education. In addition, this course prepares students to use technology to improve professional productivity and effectiveness in areas like data analysis and data representations.